MODULE 2: LASERS & OPTICAL FIBERS

PART 1: LASERS

The word LASER stands for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation. As the name itself indicates it is basically an opto-electronic device which amplifies the light.

LASER is one of the most interesting inventions of the 20^{th} century in the field of applied optics. The first ever known laser is ruby laser, developed by T.H. Maiman of USA in 1960.

IMPORTANT CHARACTERISTICS OF LASER:

- High directionality (unidirectional)
- High mono chromaticity (single wavelength)
- High coherence
- High intensity
- High focussability

Though the first laser system was built in 1960, the theory behind laser action was established by Albert Einstein in 1917.

Production of laser light is a particular consequence of interaction of radiation with matter. To understand the basic principle involved in the production of laser light, it is essential to know how an electromagnetic radiation (light) interacts with matter.

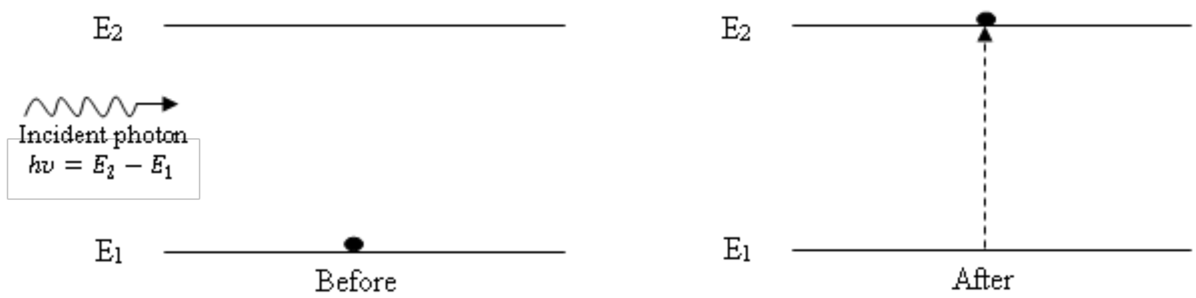
INTERACTION OF RADIATION WITH MATTER:

There are three possible ways by which radiation interacts with matter. They are

1. Induced absorption
2. Spontaneous emission
3. Stimulated emission

1. INDUCED ABSORPTION:

Induced absorption is the process of absorption of the incident photon by an atom as a result of which it makes transition from a lower energy state to higher energy state, whose energy difference is equal to that of incident photon energy.

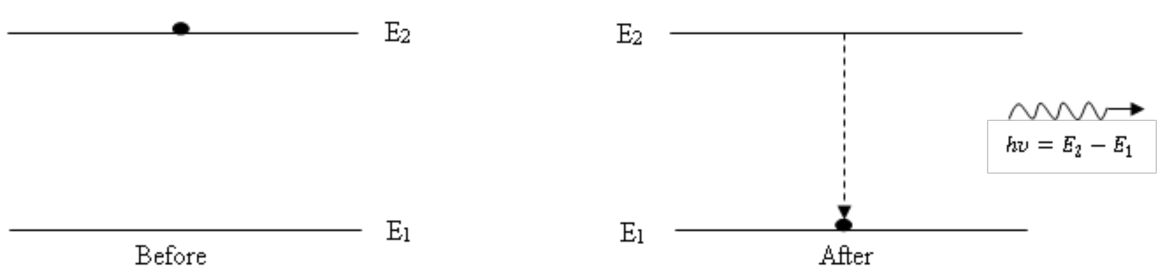


Consider an atom in its ground state (lower energy state) whose energy is E_1. Let E_2 be the energy of one of the excited states (higher energy state). Consider an electromagnetic radiation (photon) of frequency υ, such that $h\upsilon = E_2 - E_1$ is incident on this atom. The incident photon gets absorbed by the atom. As a result of it, the atom gets excited and makes transition to the higher energy state E_2 as shown in figure. This process is called induced absorption and can be represented symbolically as follows;

$$\text{Atom} + \text{Photon} \longrightarrow \text{Atom}^*$$

2. SPONTANEOUS EMISSION

Spontaneous emission is the process in which a photon is emitted when an atom in the excited state makes transition to a lower energy state without the aid of any external agency.

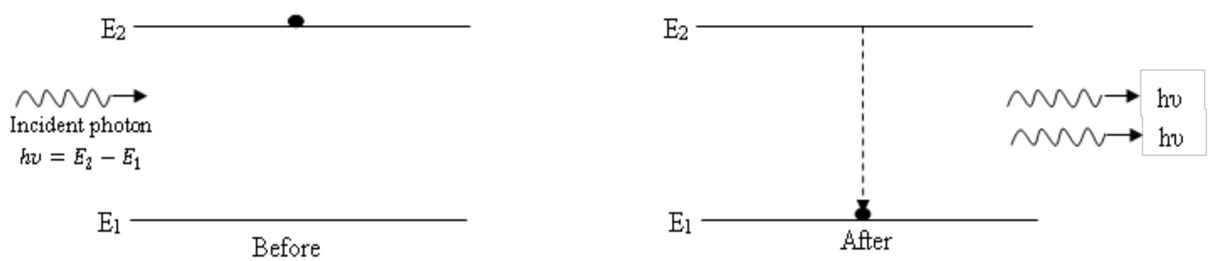


Consider an atom in the higher energy state of energy E_2. As this atom is highly unstable, gets de-excited to a lower energy state of energy E_1 without being triggered by any external influence within a short interval of time 10^{-8} - 10^{-9} s as shown in figure and emits a photon of energy $h\upsilon = E_2 - E_1$. Since the atom emits the photon without being aided by any external agency, it is called spontaneous emission. This process can be represented symbolically as follows;

$$\text{Atom}^* \longrightarrow \text{Atom} + \text{Photon}$$

3. STIMULATED EMISSION

Stimulated emission is the process of emission of a photon by an atom under the influence of a passing photon of suitable energy, due to which the atom makes the transition from a higher energy level to a lower energy state.



Consider an atom in the higher energy state of energy E_2. Let a photon having energy hυ, such that $hv = E_2 - E_1$ (where E_1 is the energy of a lower level) interacts with atom by passing in its vicinity. Under such stimulation, the atom emits a photon and makes transition to the lower energy state E_1. The emitted photon is in the same direction and has the same energy and phase as that of the incident photon and thus they are coherent. Since the emission of photon is triggered by an external influence it is called stimulated emission and is represented symbolically as follows;

$$\text{Atom}^* + \text{Photon} \longrightarrow \text{Atom} + 2\ \text{Photon}$$

Note: Population of energy levels – Boltzmann factor:

The number of atoms per unit volume present in a particular energy level at thermal equilibrium is called its population. The relation between the population of different energy levels of any physical system at thermal equilibrium was given by Boltzmann and is called Boltzmann's factor.

Let us consider two energy levels E_1 and E_2 ($E_2 > E_1$) with populations N_1 and N_2 respectively.

Then the Boltzmann factor is given by

$$\frac{N_2}{N_1} = e^{-\frac{(E_2 - E_1)}{kT}} \quad \text{or} \quad N_2 = N_1 e^{-\frac{(E_2 - E_1)}{kT}}$$

where k is the Boltzmann constant. Since $E_2 > E_1$ => $N_2 < N_1$

Hence, for a system in thermal equilibrium, the population of a higher energy state is always less than that of lower energy state.

EXPRESSION FOR RADIANT ENERGY DENSITY AT THERMAL EQUILIBRIUM IN TERMS OF EINSTEIN'S COEFFICIENTS:

Consider an atomic system at an absolute temperature T K under thermal equilibrium conditions. Let N_1 and N_2 be the number of atoms per unit volume in the energy states of energies E_1 and E_2 respectively. Let an electromagnetic radiation of frequency υ and energy density U_υ be incident on the atomic system. This photon may cause induced absorption or stimulated emission depending on the state of the particular atom on which it falls or the atom may undergo spontaneous emission.

Now consider the probability of these three processes taking place.

Induced absorption:

The rate of induced absorption depends on the number density of the lower energy state (N_1) and energy density of the incident radiation (U_υ)

$\therefore$ rate of induced absorption α $N_1 U_\upsilon$

$= B_{12} N_1 U_\upsilon$ ----------- (1)

where B_{12} is the probability constant called Einstein's co-efficient of induced absorption.

Spontaneous emission:

The rate of spontaneous emission depends on only on the number density of the higher energy state (N_2)

$\therefore$ rate of spontaneous emission α N_2

$= A_{21} N_2$ ----------- (2)

where A_{21} is the probability constant called Einstein's co-efficient of spontaneous emission.

Stimulated emission:

The rate of stimulated emission depends on the number density of the higher energy state (N_2) and energy density of the incident radiation (U_υ)

$\therefore$ rate of stimulated emission α $N_2 U_\upsilon$

$= B_{21} N_2 U_\upsilon$ ----------- (3)

where B_{12} is the probability constant called Einstein's co-efficient of stimulated emission.
According to Einstein's assumption, at thermal equilibrium rate of upward transition is equal to rate of downward transition i.e. the rate of induced absorption must be equal to the sum of the rates of spontaneous and stimulated emissions.

Rate of induced absorption = Rate of spontaneous emission + Rate of stimulated emission ….(4)

Using (1), (2) and (3) in (4) we get

$$B_{12} U_\nu N_1 = A_{21} N_2 + B_{21} U_\nu N_2$$

$$U_\nu [B_{12} N_1 - B_{21} N_2] = A_{21} N_2$$

$$U_\upsilon = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2}$$

By rearranging the above equation, we get

$$U_\upsilon = \frac{A_{21}}{B_{21}}\left[\frac{1}{\frac{B_{12}N_1}{B_{21}N_2} - 1}\right] \quad ------\rightarrow (5)$$

From Boltzmann's relation we have

$$\frac{N_2}{N_1} = e^{-\frac{(E_2-E_1)}{kT}} \text{ or } \frac{N_2}{N_1} = e^{-\frac{h\upsilon}{kT}} \qquad \because h\upsilon = E_2 - E_1$$

$$\therefore \quad \frac{N_1}{N_2} = e^{\frac{h\upsilon}{kT}}$$

∴ equation (5) becomes

$$U_\upsilon = \frac{A_{21}}{B_{21}}\left[\frac{1}{\frac{B_{12}}{B_{21}} e^{\frac{h\upsilon}{kT}} - 1}\right] \quad ------\rightarrow (6)$$

According to Planck's law of radiation

$$U_\upsilon = \frac{8\pi h\upsilon^3}{c^3}\left[\frac{1}{e^{\frac{h\upsilon}{kT}} - 1}\right] \quad ------\rightarrow (7)$$

Comparing equations (6) and (7) we get

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\upsilon^3}{c^3} \qquad and \qquad \frac{B_{12}}{B_{21}} = 1$$

Thus, taking $A_{21} = A$ and $B_{21} = B_{12} = B$

Equation (6) reduces to

$$U_\upsilon = \frac{A}{B}\left[\frac{1}{e^{\frac{h\upsilon}{kT}} - 1}\right]$$

This is the expression for energy density in terms of Einstein's coefficients A and B.

***Note*: -** According to Einstein's theory

$$\frac{Rate\ of\ stimulated\ emission}{Rate\ of\ induced\ absorption} = \frac{B_{21}U_{\upsilon}N_2}{B_{12}U_{\upsilon}N_1} = \frac{N_2}{N_1} \qquad \because\ B_{12} = B_{21}$$

From the above ratio, it is clear that, stimulated emission dominates over induced absorption and hence leads to laser action only when $N_2 > N_1$. In order to achieve laser action, the system must be in population inversion condition.

CONDITIONS FOR LASER ACTION:

For continuous laser beam emission, the following two conditions must be satisfied.

1. Population Inversion

Population inversion is the condition in which the number of atoms in the higher energy state is greater than that of a lower energy state.

According to Boltzmann's relation, under normal conditions, in an atomic system, the number density of atoms in the lower energy state is greater than higher energy state. On supplying the energy to the system, the number of atoms in the ground state decreases and that of higher energy state increases, after some time, a situation is reached in which number of atoms in the higher energy state will be greater than that of lower energy state. This condition of inverted population is called as population inversion.

***Need for population inversion*:**

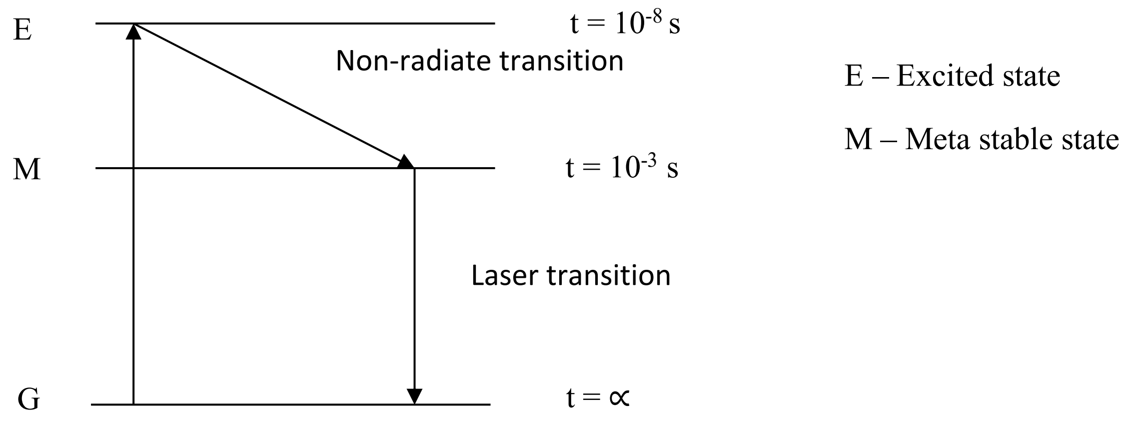
The population inversion is needed in order to enhance the rate of stimulated emission and hence to achieve light amplification.

Note: - The population inversion can be achieved by a technique called pumping in which energy is supplied to system resulting in pumping of atoms from lower energy state to a higher energy state. There are different pumping techniques namely Optical pumping (light energy is supplied, ex: ruby laser), Electrical pumping (Electrical energy is supplied, ex: semiconductor laser) etc.

2. Metastable state:

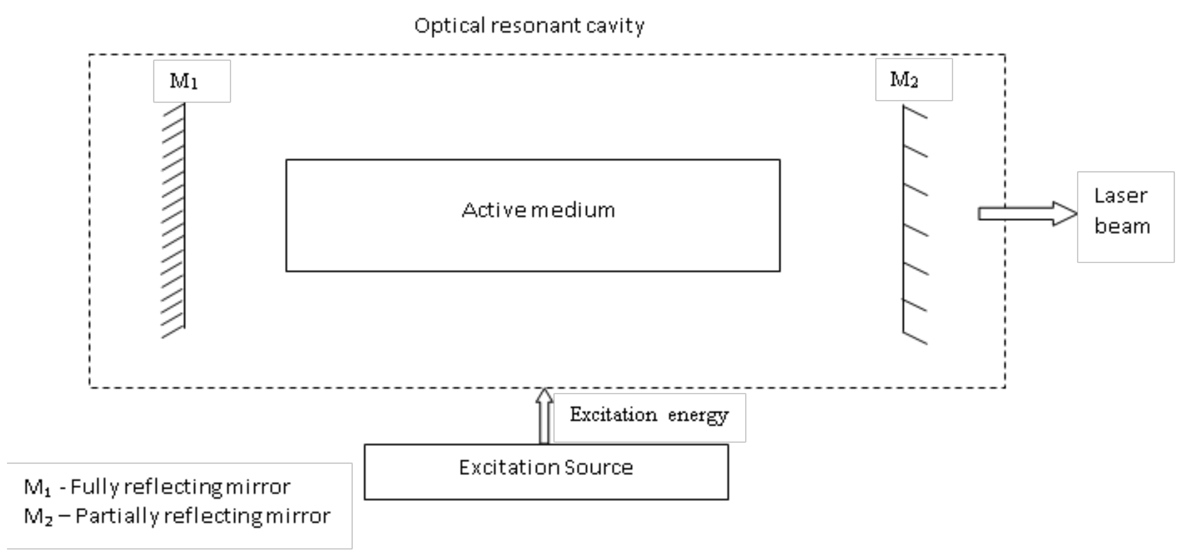
Metastable state is the energy state in a system having longer lifetime (10^{-3}s to10^{-2} s) and helps in achieving population inversion.

When atoms are pumped from ground state to excited states they stay there over a very short time (10^{-8} s – 10^{-9} s) and return to their lower energy states. Hence in an atomic system under normal conditions, the state of inverted population does not exist. However, it is possible to achieve the population inversion condition in certain systems which possess a special kind of excited energy levels having longer life time ($\approx 10^{-3}$ s to 10^{-2}) called meta stable state.



Requisites of a laser system:

The important requisites of a laser system are



Sketch of typical laser system

1. An excitation source for pumping action (Energy source)
2. An active medium which supports population inversion
3. An optical resonant cavity / laser cavity which provide optical resonance

Excitation source:

In order to achieve laser action system must be in population inversion condition, to achieve this i.e., for pumping action a source of energy is required. This source is called excitation source.

Example: In He – Ne laser and semiconductor laser, “power supply” is the excitation source.

In Ruby laser, “Xenon flash lamp” is the excitation source.

In carbon dioxide laser “power supply” is the excitation source

Active medium:

The quantum system between whose energy levels, the pumping and laser action takes place is called an active medium. In a lasing system consisting of a large number of atoms of different kind, only a small fraction of atoms of a particular species are responsible for stimulated emission and consequent light amplification. They are called active centers. The remaining bulk of the medium acts the role of host and supports the active centers.

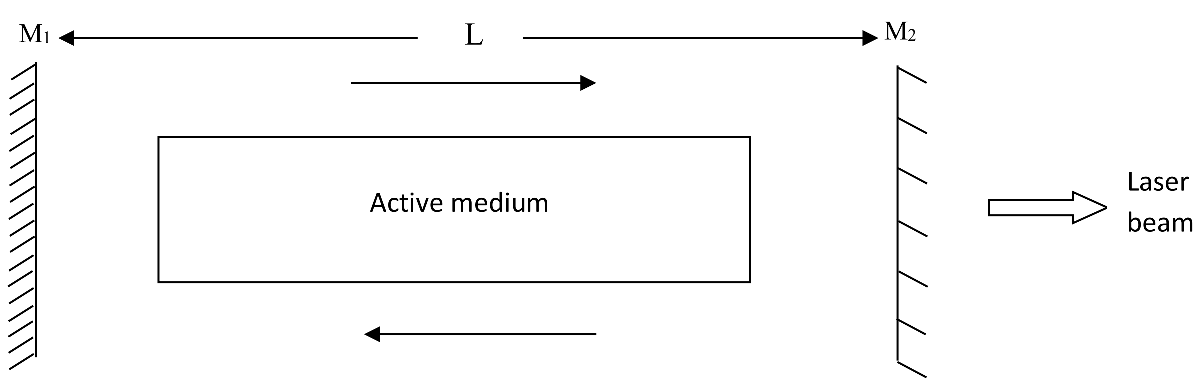
Examples: In He – Ne laser, Ne – atoms are active.

In Ruby laser, Chromium ions are active.

In carbon dioxide laser CO_2 molecules are active.

Optical resonant cavity [laser cavity]:

A laser device consists of an active medium bound between two mirrors; one is partially reflecting while the other is fully reflecting. The two reflecting mirrors along with the active medium form a cavity called as optical resonant cavity or laser cavity.



M_1 - Fully reflecting mirror

M_2 – Partially reflecting mirror

Inside the laser cavity, the photons travel back and forth through the active medium, due to multiple reflections at the mirrors, the energy density of the photons increases. As a consequence, there will be two waves setup one moving to the right and the other to the left. For amplification of light, these two waves must interfere constructively.

In order to arrange for constructive interference, the distance between the mirrors should be such that, path difference between the interfering waves must be equal to integral multiple of the wavelength.

$$\text{i. e.} \quad \text{path difference} = n\lambda$$

$$2L = n\lambda$$

or

$$\boxed{L = \frac{n\lambda}{2}} \longrightarrow \text{Resonance condition}$$

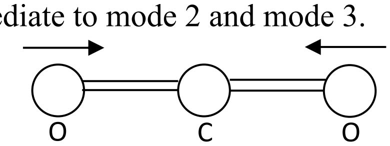
Where n = 1, 2, 3… And λ is the wavelength of the laser beam inside the material of the active medium. In such case a standing wave is established within the cavity and the cavity is said to be at "resonance".

Vibrational Energy states of Carbon dioxide:

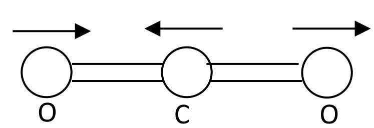
Carbon dioxide laser emits laser radiation due to transition between vibrational states of the same electronic state. These vibrational states are associated with the vibrational modes of carbon dioxide.

In carbon dioxide molecule there are three fundamental modes of vibration:

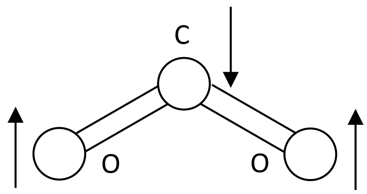
1. Symmetric stretching mode: In this mode, the carbon atom is stationary and the oxygen atoms oscillate simultaneously to and fro along the molecular axis. Energy corresponding to this mode is intermediate to mode 2 and mode 3.



2. Asymmetric stretching mode: In this mode, all three atoms oscillate along the molecular axis in such a way that oxygen atoms move opposite to the carbon atom. Energy corresponding to this mode is highest.



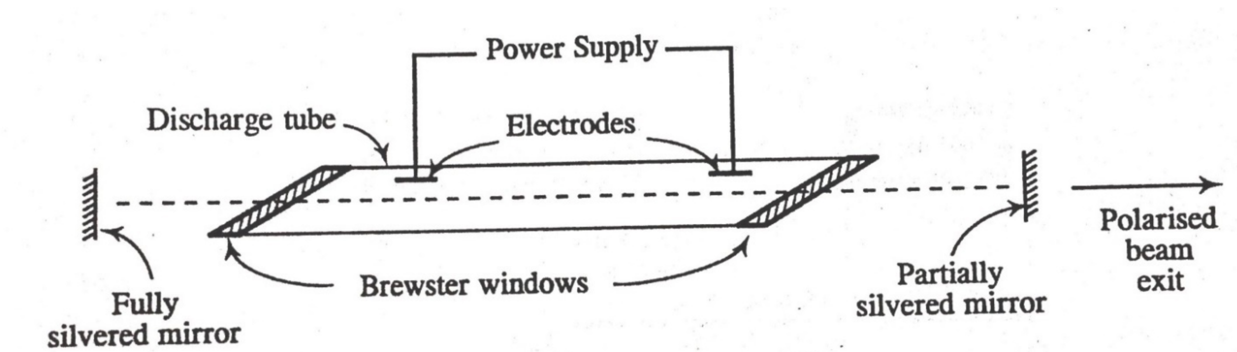
3. Bending mode: In this mode, oxygen atoms and carbon atom move perpendicular to the molecular axis. Energies corresponding to this mode are less than mode 1 & mode 2.



Construction and working of Carbon Dioxide Laser:

The **carbon dioxide laser** (**CO_2 laser**) was developed by C K N Patel when he was working in Bell Labs, USA. The carbon dioxide gas laser is capable of producing continuous output powers above 10 kilowatts. It is a four-level molecular laser and operates at 10.6 µm and 9.6 µm in far IR region.

Construction:

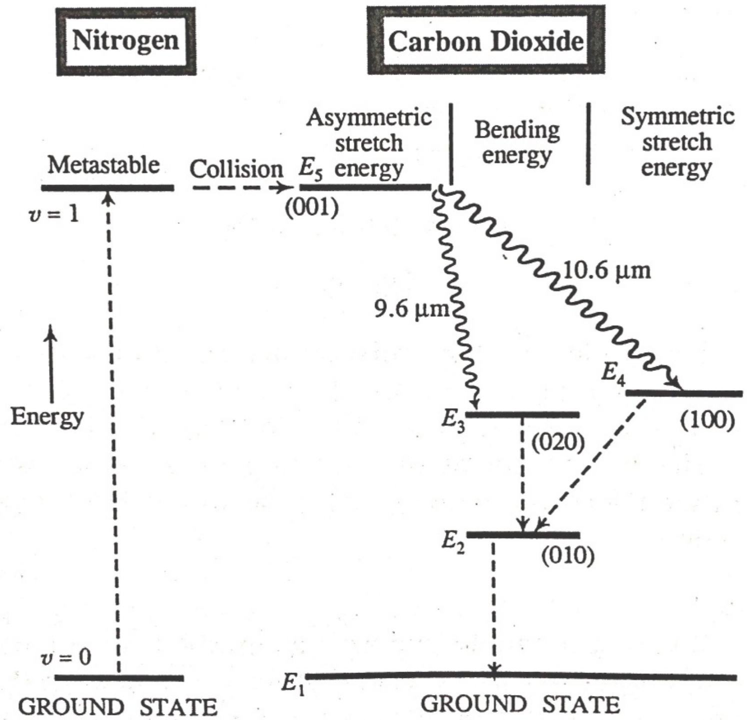


CARBON DIOXIDE LASER

- CO_2 laser consists of a discharge tube of 2.5cm in diameter and 5cm length.
- The tube is filled with a mixture of CO_2, N_2 and He gases in the ratio of 1:2:3.
- The tube is provided with electrodes for electrical discharge.
- Brewster's windows are attached to tube to get plane polarized light from M_2
- Two plane mirrors M_1 and M_2 are fixed on either side of the tube, where M_1 is fully silvered and M_2 is partially silvered to act as laser cavity.
- The pressure inside the tube is 6-17torr.

Working:

The internal vibrations of carbon dioxide molecule are the linear combination of these three modes. The energy level diagram of carbon dioxide laser is as shown in below figure.



CO_2 LASER ENERGY LEVEL DIAGRAM

- Suitable voltage is applied across the two electrodes due to which electrical discharge is initiated in the tube.
- During discharge, many free electrons are free from the gas molecules. These free electrons collide with the N_2 and CO_2 molecules in their path.
- There are two possible kinds of interactions, which is collision of I kind.

$$e_1 + N_2 = e_2 + N_2*$$
$$e_1 + \mathrm{CO}_2 = e_2 + \mathrm{CO}_2*$$

i.e. N_2 and CO_2 molecules are raised to V=1 and E_5 energy levels respectively.

- There by CO_2 molecules are raised to E_5 energy level and N_2 molecules are raised to V=1 level which is the meta-stable state and its population builds up.
- There is a close coincidence in energy of E_5 state of CO_2 molecule with V=1 state of N_2 molecule. Therefore, there will be a resonant transfer takes place, which is collision of II kind.

$$N_2^* + CO_2 = N_2 + CO_2^*$$

- Due to this population of E5 i.e. (001) level rapidly increases which leads to population inversion.

- Once the population inversion is established between E_5 & E_4 level and between E_5 & E_3 level laser transition starts.
- Transition from $E_5 \rightarrow E_4$ $gives\ radiation\ of\ wavelength$ $10.6\mu m$
  Transition from $E_5 \rightarrow E_3$ $gives\ radiation\ of\ wavelength$ $9.6\mu m$
- The role of helium is to de-excite the CO_2 molecules to ground state.

Applications: CO_2 lasers are widely used for material processing, in particular for

- Cutting plastic materials, wood, die boards, etc.
- Cutting and welding metals such as stainless steel, aluminum or copper, applying multi-kilowatt powers.
- Other applications include laser surgery (including ophthalmology) and range finding.

APPLICATION OF LASERS

Holography

Holography is the technique of capturing three dimensional pictorial details of an object on a two-dimensional recoding aid, by using the phenomenon of interference.

The term holography is coined from two Greek words HOLOS and GRAPHOS, which mean complete and recording/writing respectively.

Holographic technique was first invented in the year 1948 by Dennis Gabor, a British Scientist. Gabor was awarded the Nobel Prize in physics for the year 1971.

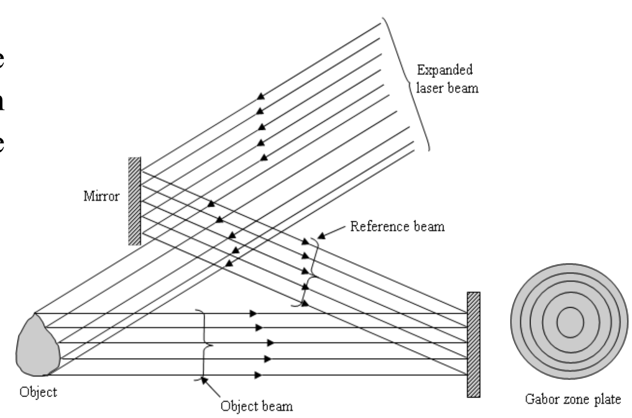
Recording (or Construction) and Reconstruction of images:

In order to record 3D pictorial details of an object in holography there are two stages namely.

- Recording or construction of image of an object
- Reconstruction of the image

Recording or construction of image of an object:

There are two methods of recording. They are wave front division technique and amplitude division technique. Generally, the wave front division technique is used for recording and is explained below.



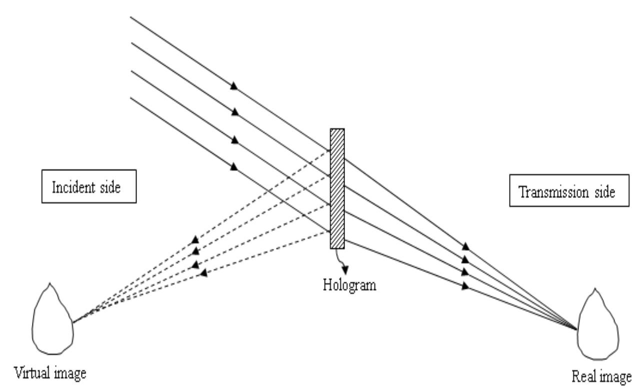
The expanded laser beam which can be obtained by using a laser system is directed towards mirror and an object as shown in figure. Part of the beam is incident on the mirror and rest passes past the mirror and incident on the object.

A photographic plate is placed in such a position that it receives the light reflected from both the mirror and the object. The part of the light reflected from the mirror and is incident on the photographic plate will be in the form of plane wave fronts. This is called the reference beam.

Let us consider the part of the light reflected from the object. Every point on the object scatters the incident light all round. Due to such a scattering, spherical wave fronts proceed from each point on the object, a part of each of which will be incident on the photographic plate. This part is called the object beam. The photographic plate responds to the resultant effect of interference between the spherical wavelets of the object beam and the plane waves of the reference beam. The interference pattern consists of concentric circular rings pattern (dark and bright). The ring pattern is called Gabor Zone Plate. Since, every set of spherical wavelets that starts from each point on the object generates its own zone plate, the recording consists of countless number of such zone plates, the center of each of which is slightly displaced from the other. The photographic plate after development is called HOLOGRAM.

Reconstruction of the image:

For the reconstruction of the image, the original laser beam (one which was used during construction) is directed at the hologram in the same direction as the reference beam was incident on it at the stage of recording as shown above figure. Upon incidence, the laser beam undergoes diffraction in the hologram. Because of diffraction, secondary wavelets originate from each constituent zone plate which interfere constructively in certain directions and generate both a real and a virtual image of the corresponding point of the object.



Since the beam is incident on the entire hologram each and every zone plate participates in this process to generate both real and virtual image of the object point by point. By looking through the hologram from the transmission side, it appears as though the original object is lying on the other side of it at the same place. This is the virtual image due to reconstruction. By switching the direction of view different set of points which correspond to construction interference are observed, which regenerate a different perspective of the object. Thus, the sense of depth is felt.

The other wave stream starting from the various zone plates is convergent and gives rise to the real image of the object. This image can be photographed.

PART 2: OPTICAL FIBERS:

An optical fiber is a cylinder of transparent dielectric material of variable refractive index and designed to guide visible and infrared light over long distances using the phenomenon of total internal reflection.

Construction of optical fiber:

A typical optical fiber consists of two parts, namely

1. The inner part called Core, a thin cylinder which is made of transparent dielectric material like high purity silica glass or plastic with higher refractive index.
2. The outer part called Cladding, a hollow cylinder surrounding the core, which is also made of same material as that of core but with lesser refractive index. There is a material continuity between core and cladding & the combination of core and the cladding is enclosed in a protective layer made of polyethylene or polyvinylchloride (PVC) jacket called sheath. The cross-sectional view of an optical fiber is as shown in figure1.

The core is the transfer medium; cladding reflects the light beam inward, while the sheathing protects the fiber from the environmental effects (such as abrasions, contamination and moisture) and also increases the mechanical strength of the fiber. A typical optical fiber is as shown in figure 2.

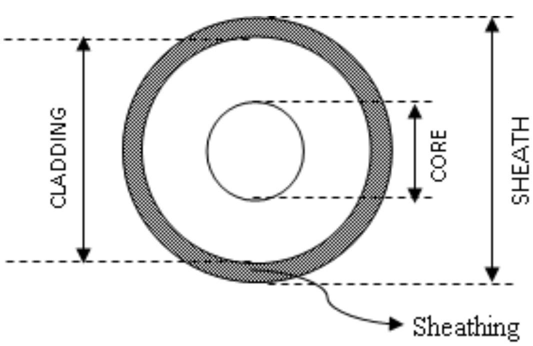


Fig. 1. Cross sectional view

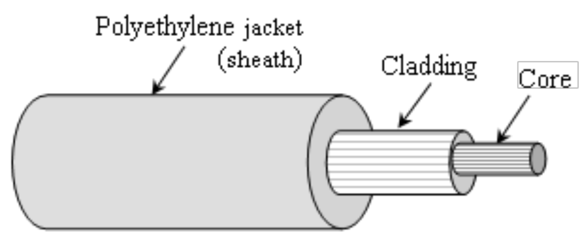


Fig. 2. 3D view

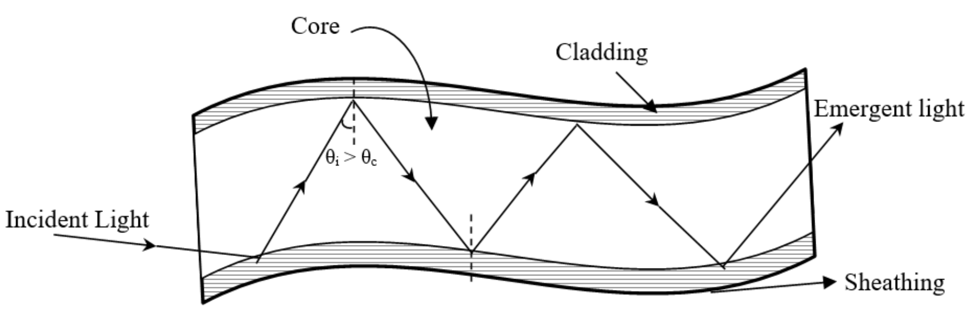
Propagation of Light in Optical Fibers:

Principle:

The basic principle of transmission of light through an optical fiber is the phenomenon of total internal reflection. In case of T I R, there is absolutely no loss of light energy, at the reflecting surface since entire incident light energy is returned along the reflected light without any absorption or transmission. Hence the phenomenon is called as total internal reflection.

Propagation mechanism:

When a ray of light enters into the core [medium of higher refractive index] at an angle of incidence greater than critical angle, it is completely reflected back into the core medium at core – cladding interface without loss of energy. Under such circumstances, the incident light beam is completely trapped inside the core of an optical fiber and cannot escape from it. As the light travels inside the core, it starts undergoing multiple reflections at the core – cladding interface and each reflection will be perfect. As a result of it, light will travel in a direction parallel to the axis of the core in zig - zag fashion without loss until it reaches the other end of the fiber as shown in figure 4. Since, an optical fiber guides a beam of light through it; it is termed as light



guide or optical waveguide.

Figure 4: Propagation mechanism of light in optical fibers

Acceptance angle (θ_0):

Acceptance angle is defined as the maximum angle of incidence of the light beam at the centre of the core up to which total internal reflection takes place at the core – cladding interface.

Numerical Aperture (NA):

The numerical aperture (NA) of an optical fiber is defined as the product of refractive index of surrounding medium (n_0) and sine of the acceptance angle.

$$i.e. \quad \mathbf{NA} = n_0 \sin\theta_0$$

If the fiber is surrounded by air, then $n_0 = 1$, then

$$\mathbf{NA} = \sin\theta_0$$

Significance of NA:

The numerical aperture of an optical fiber signifies the light gathering capacity of an optical fiber i.e. amount of light can be accepted by the fiber.

Expression for Acceptance angle and Numerical Aperture (Condition for Propagation of Light in an Optical Fiber):

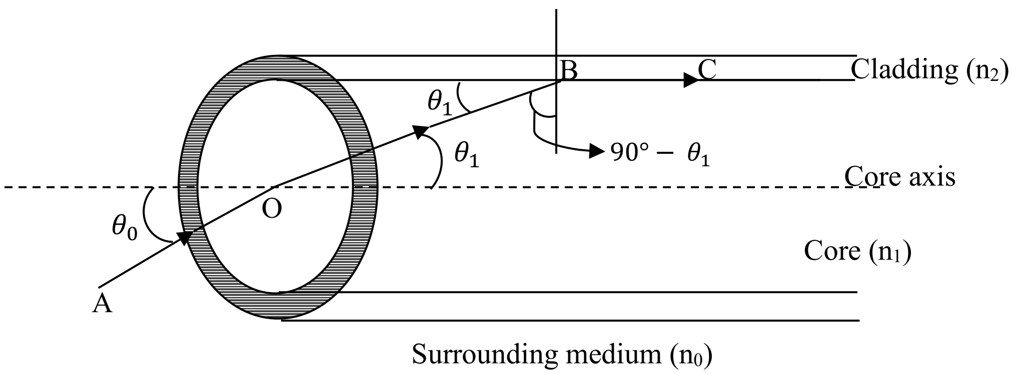


Figure 6: Propagation of light in optical fibers

Let n_0, n_1 and n_2 be the refractive indices of the surrounding medium, core and cladding of the optical fiber respectively. Consider a ray of light AO entering into the core at an angle θ_0 to the core axis of the fiber. Let it be refracted along OB at an angle θ_1 in the core and further proceed to fall at critical angle ($\theta_c = 90 - \theta_1$) for the core – cladding interface at B. Since the ray falls on the interface at critical angle, the ray is refracted at 90^o to the normal drawn to the interface i.e. it grazes the interface along BC as shown in figure 6.

Now for refraction at the point of entry of the ray AO into the core,

Apply Snell's law at 'O' for the first pair of media (the surrounding medium of refractive index n_0 and the core medium of refractive index n_1) with angle of incidence $\theta_0$and angle of refraction θ_1

$$n_0 \sin\theta_0 = n_1 \sin\theta_1$$

or
$$\sin\theta_0 = \frac{n_1}{n_0}\sin\theta_1 \qquad \text{--------} \blacktriangleright (1)$$

Now for refraction at the point B on the interface, here the refracted ray for the first pair of media will be the incident ray for the next pair of media.

Again apply Snell's law at B for the next pair of media (the core medium of refractive index n_1 and the cladding medium of refractive index n_2) with the angle of incidence $90 - \theta_1$ and angle of refraction 90^0, we get

$$n_1 \sin(90 - \theta_1) = n_2 \sin 90°$$

$$n_1 \cos\theta_1 = n_2$$

or $$\cos\theta_1 = \frac{n_2}{n_1} \qquad \text{--------} \blacktriangleright (2)$$

Using the mathematical identity $\sin^2\theta + cos^2\theta = 1$, equation (1) can be written as

$$\sin\theta_0 = \frac{n_1}{n_0}\sqrt{1-\cos^2\theta_1} \qquad \text{--------} \blacktriangleright (3)$$

Substitute equation (2) in equation (3) we get,

$$= \frac{n_1}{n_0}\sqrt{1-\left(\frac{n_2}{n_1}\right)^2}$$

$$= \frac{n_1}{n_0}\sqrt{\frac{n_1^2-n_2^2}{n_1^2}}$$

i.e. $$\sin\theta_0 = \frac{\sqrt{n_1^2-n_2^2}}{n_0}$$

$$\boxed{\theta_0 = \sin^{-1}\left(\frac{\sqrt{n_1^2-n_2^2}}{n_0}\right)} \qquad \text{--------} \blacktriangleright (4)$$

This is the expression for acceptance angle.

From equation (4), it is clear that for a given optical fiber, the acceptance angle varies inversely as the refractive index of the surrounding medium i.e acceptance angle is more for a given optical fiber if it is operated in a surrounding medium of lower refractive index or vice-versa.

Equation (4) can be written as

$$n_0\sin\theta_0 = \sqrt{n_1^2-n_2^2}$$

But $n_0\sin\theta_0 = NA$

Therefore, $$\boxed{NA = \sqrt{n_1^2-n_2^2}} \qquad \text{--------} \blacktriangleright (5)$$

This is the expression for numerical aperture.

From equation (5), it is evident that the NA depends only on the refractive indices of the core and cladding material of the fiber and is independent of refractive index surrounding medium and fiber dimensions.

Condition for propagation of light in an optical fiber:

In air medium, if θ_i is the angle of incidence of an incident ray at the center of core, then the ray will be accepted and transmitted through the fiber if $\theta_i < \theta_0$.

i.e. $\sin\theta_i < \sin\theta_0$

or $\sin\theta_i < NA$

This is the condition for propagation of light through the fiber i.e. sine of the angle of incidence of the incident ray at the centre of the core should be less than the numerical aperture of the fiber.

Fractional index change (Δ):

The fractional index change (Δ) is defined as the ratio of difference between the refractive indices of the core and cladding to the refractive index of the core of an optical fiber.

If n_1 and n_2 are the refractive indices of the core and cladding respectively, then fractional index change Δ is given by

$$\Delta = \frac{n_1 - n_2}{n_1}$$

Note:

1. Δ is also called "relative refractive index difference".
2. Δ is always positive because $n_1 > n_2$ for total internal reflection.

Modes of Propagation:

The term Modes refers to the number of paths available in the optical fiber for the propagation of light through it. When a beam of light containing in a number of rays (waves) enters into an optical fiber, only certain selected number of rays (modes) will be sustained for propagation in the fiber.

If only one ray is sustained or one path is available for propagation, corresponding mode of propagation is called **single mode** and if it takes many paths through the fiber, it is called **multimode**.

V – Number (V - Parameter):

The number of modes of light that can be guided by an optical fiber depends on the core size, refractive indices of the core and cladding materials and operating wavelength. It can be mathematically defined using a normalized frequency parameter called V – Parameter or V–number and is denoted simply by V.

If'd' is the diameter of the core, n_1 and n_2 are the refractive indices of the core and cladding respectively and λ is the operating wavelength and the fiber is surrounded by air, then V – Number is given by

$$\boxed{V = \frac{\pi d}{\lambda}(NA)} \quad \text{or} \quad \boxed{V = \frac{\pi d}{\lambda}\sqrt{n_1^2 - n_2^2}} \quad \because NA = \sqrt{n_1^2 - n_2^2} \quad for\ air$$

Note:

1. For V >>1, the number of modes (N) supported by and optical fiber is given by

$$\boxed{N \approx \frac{V^2}{2} = \frac{1}{2}\left[\frac{\pi d}{\lambda} NA\right]^2}$$

2. For Single mode optical fiber V ≤ 2.405 and for a multimode optical fiber V > 2.405.

Refractive Index Profile:

The variation of refractive index of the fiber material with respect to the radial distance from the axis of the fiber is represented by a curve called refractive index profile. There are two types of refractive index profile namely **Step index** refractive index profile and **Graded index** refractive index profile.

In case of Step index profile the refractive index of core is constant, where as in Graded index profile refractive index of core is not constant, but varies as a function of radial distance from the center of the core. The refractive index is maximum at the center of the core and decreases gradually in radially outward direction and becomes equal to that of the cladding at the interface as shown in figures 7(a) and 7(b).

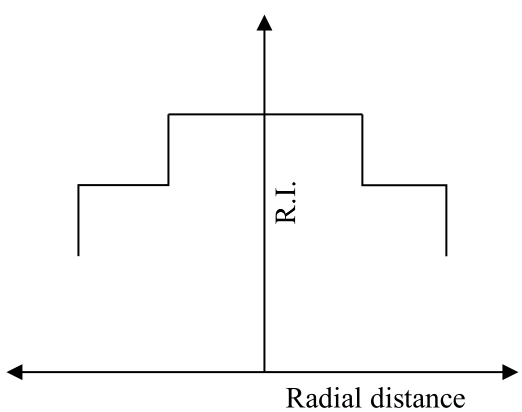


Figure 7(a): Step Index

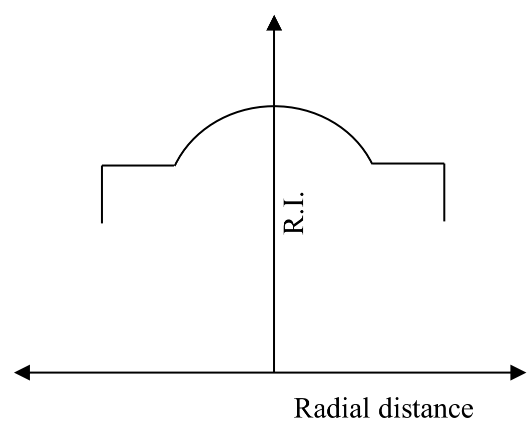


Figure 7(b): Graded Index

Types of Optical Fibers:

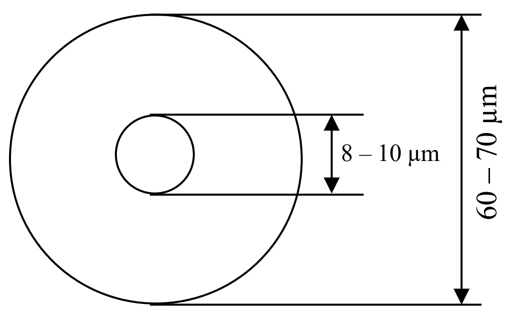
Depending on the refractive index profile and the number of modes that fiber can guide, optical fibers are classified into three types, namely;

1. Step Index Single Mode Fiber (or) Single Mode Fiber
2. Step Index Multimode Fiber
3. Graded Index Multimode Fiber

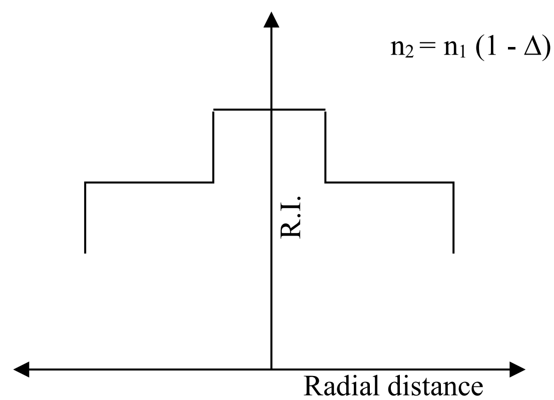
1. Single Mode Fiber

The geometry, refractive index profile and ray propagation for single mode fiber are shown in below Figure 8. A single mode fiber consists of very thin core material (8 – 10 µm in diameter) of uniform refractive index value. The core is surrounded by a thick cladding material (of external diameter 60 to 70 µm) of uniform refractive index but of lesser value compared to that of core. This results in a sudden decrease in the value of refractive index at the core – cladding interface. Thus, its refractive index profile takes the shape of step. Because of its narrow core, it can guide only one mode of propagation.

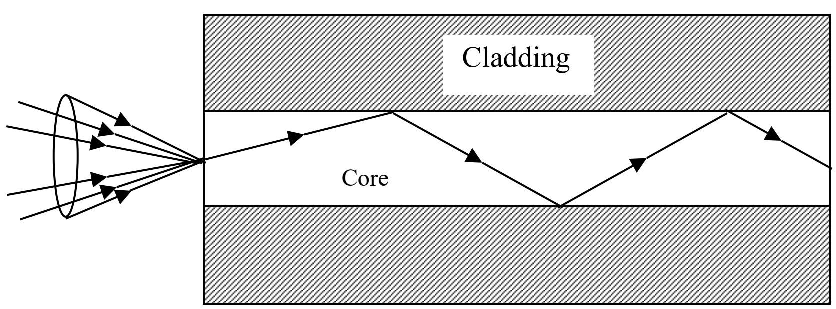
Uses: The single mode optical fibers are most suitable for long distance, high data rate communication and hence find applications in long distance communication and submarine cable systems.



Geometry



Refractive index profile



Ray propagation

Figure 8: Schematic representation of single mode fiber

2. Step Index Multimode Fiber

The geometry, refractive index profile and ray propagation for step index multimode mode fiber are shown in below figure 9. A step index multimode fiber is very much similar in its construction to that of single mode fiber except its core is of large diameter, by virtue of which it can able to support a large number of modes of propagation.

A typical step index multimode fiber consists of a thick core (of diameter 50 – 200 μm). The core is surrounded by a thin cladding material (of diameter 100 – 250 μm). Its refractive index profile is also similar to that of single mode fiber but with larger plane region for the core.

Uses: Step index multimode fibers are most suitable for low band width and short distance communications and find application in data links which has low band width requirements.

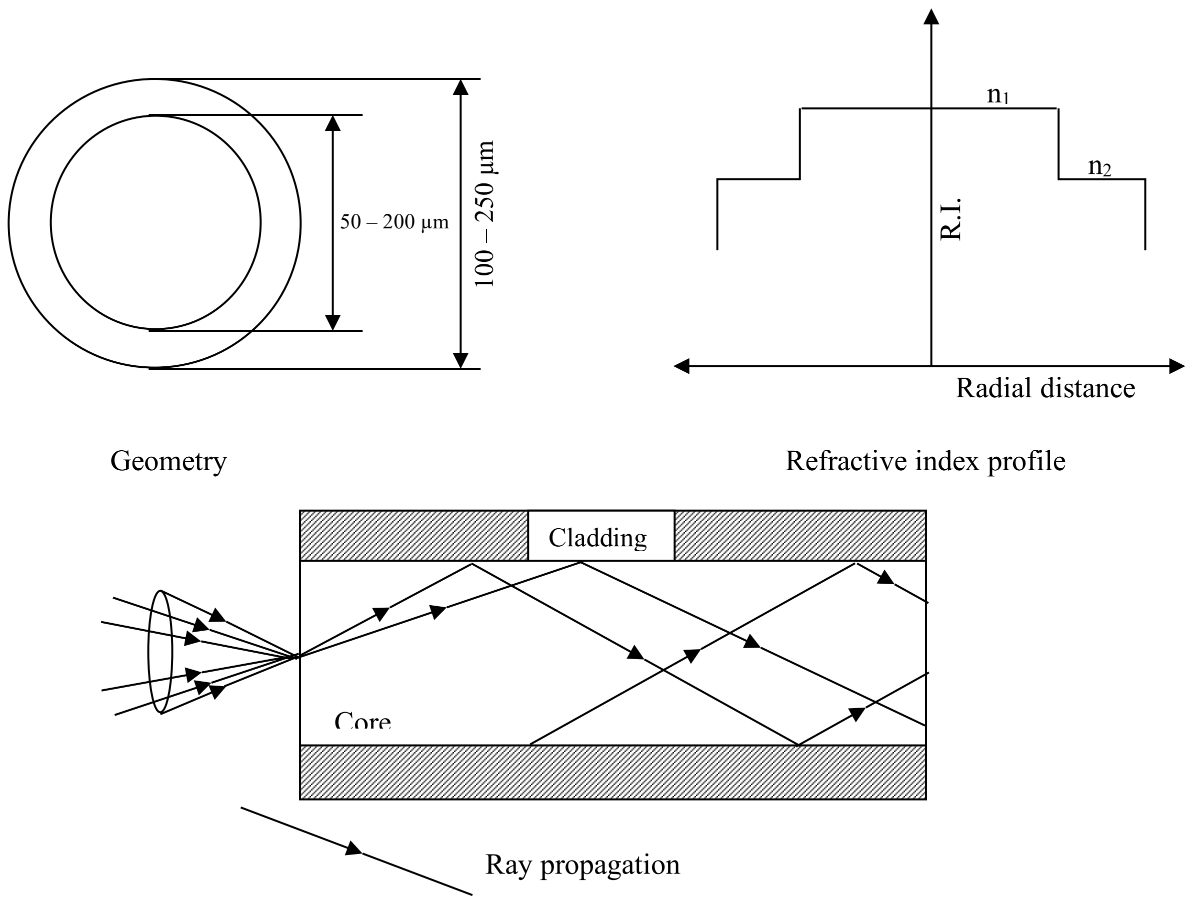


Figure 9: Schematic representation of Step index multimode fiber

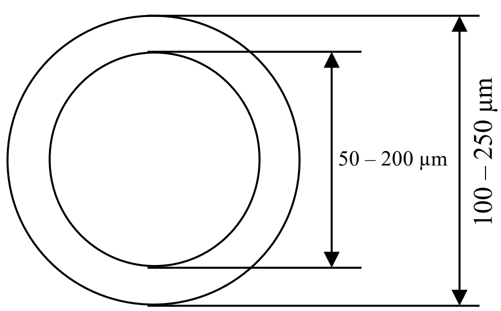
3. Graded Index Multimode Fiber

The geometry, refractive index profile and ray propagation for graded index multimode mode fiber are shown in figure 10.

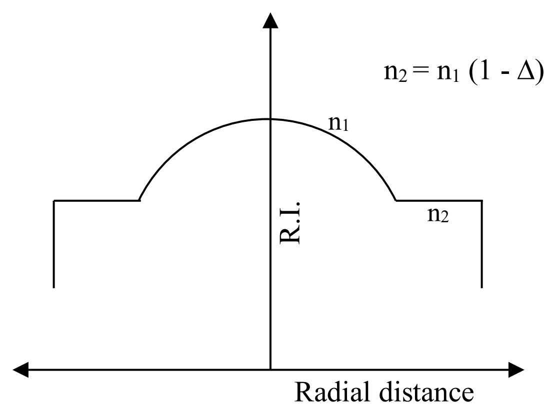
Graded index multimode fiber is also denoted as GRIN multimode fibers. A "GRIN" multimode fiber has the same geometry as that of step index multimode fiber.

A typical GRIN multimode fiber consists of a core material (of diameter 50 – 200 μm) made of concentric layers of different refractive indices. i.e. the refractive index of the core is not uniform, it decreases in the radially outward direction from the axis and becomes equal to that of the cladding at the interface. The cladding (of external diameter 100 – 250 μm) material is made of uniform refractive index. Such a profile causes a periodic focusing of the different mode of wave propagation through the fiber.

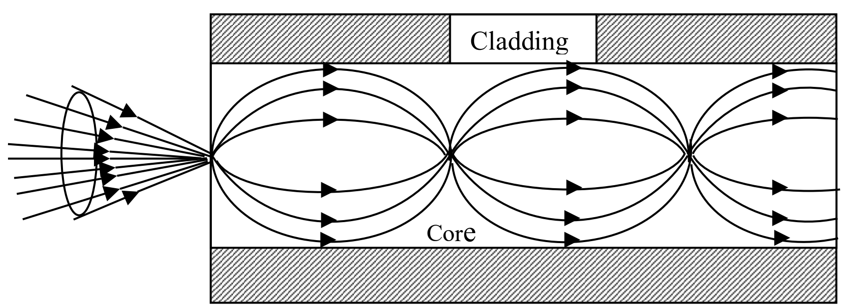
Uses: GRIN multimode fibers are most suitable for large band width, medium distance and medium data rate communication systems and find application in the telephone trunk between central offices.



Geometry



Refractive index profile



Ray propagation

Figure 10: Schematic representation of Graded index multimode fiber

Attenuation or Fiber Loss:

The loss of power suffered by the optical signal as it propagates through the fiber is called Attenuation. It is also called the fiber loss.

The net attenuation is measured by a factor called attenuation co-efficient (α) in dB/km and is given by the relation

$$\boxed{\alpha = -\frac{10}{L}\log\left(\frac{P_{out}}{P_{in}}\right) \qquad \ldots dB/km}$$

where P_{in} - Power coupled into the fiber & P_{out} - Power output at the end

L - Length of the fiber (in km)

Causes of attenuation or fiber loss:

There are three mechanisms through which attenuation takes place

1. Absorption Loss
2. Scattering Loss
3. Radiation Loss

1. Absorption Loss:

a) Absorption by Impurities

The impurities generally present in the fiber glass are transition metal ions such as iron chromium etc. During signal propagation photons interact with these impurities and gets absorbed which is a loss.

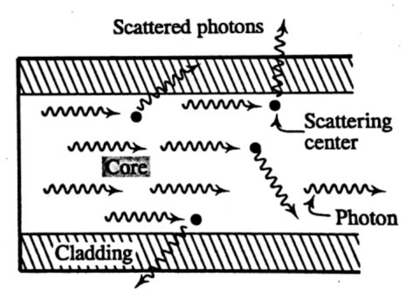
b) Intrinsic Absorption:

The fiber itself as a material has a tendency to absorb small amount of light this absorption that takes place assuming that there are no impurities in it is called intrinsic absorption.

2. Scattering Loss:

a) Rayleigh Scattering:

While the signal travels in the fiber the photons may be scattered because of sharp changes in refractive index values inside the glass over distances which are small and comparable to wavelength of light. These sharp variations are set into glass during solidification from molten state. This type of scattering is called Rayleigh scattering.



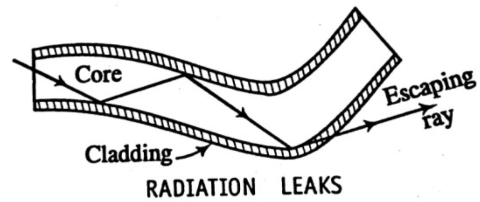
RAYLEIGH SCATTERING

b) Other Scattering:

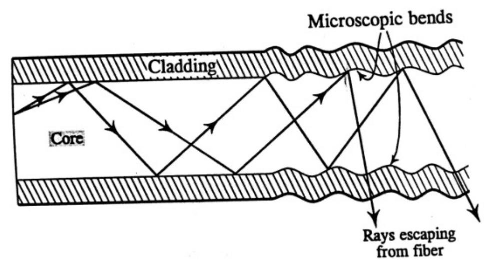
Scattering of light also takes place due to defects in the optical fiber such as trapped gas bubbles, unreacted starting materials. But the latest manufacturing methods make these losses negligible compared to Rayleigh scattering.

3. **Radiation Loss:**
   It occurs due to
   a) Macroscopic Bends
   b) Microscopic Bends

   a) **Macroscopic Bends**
   These are bends in the fiber with the radii much larger compared to fiber diameter, if the ray strikes the core cladding interface at these bends the ray escapes out of optical fiber. These bends occur while rapping the optical fiber or turning it around a corner.



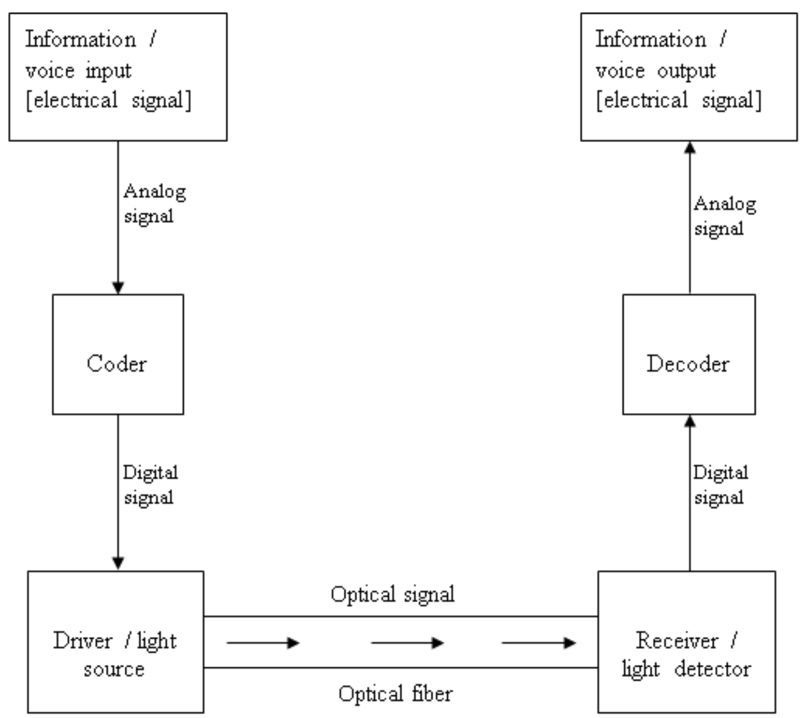
   b) **Microscopic Bends:**
   These bends are small fluctuations in linearity of the optical fiber. They occur due to non uniformity in manufacturing or by lateral pressure applied during the cabling of the fiber which causes irregular reflections and the radiation leaks through the fiber.



Application of Optical Fibers in Point-to-Point Communication:

A simple point to point optical communication system in a block diagram representation is as shown in figure 10. The optical communication system consists of three basic components.

1. An optical transmitter i.e., the driver / light source to transmit the signals.
2. The communication medium i.e., optical fiber.
3. An optical receiver, usually a photo detector to convert light pulses back into electrical signal.



The given information / voice to be transmitted will be in analog electrical signal format and this analog signal is converted to binary (digital) format by using an electrical system coder. The digital signal will be in the form of a stream of electrical pulses, these electrical pulses are converted into optical pulses by using an optical source such as LED or Laser and coupled into the optical fiber cable at an angle of incidence less than that of acceptance cone angle.

The optical pulses inside the fiber undergo total internal reflection and reach the other end of the fiber. Finally at the receiving end, the optical pulse from the fiber is fed into a photo detector where the signal is converted into electrical pulses. These pulses are fed to a decoder which converts the sequence of digital data stream into an analog signal and converted into usable format like audio / video.

Advantages of Optical Communication:

1. Large band width: Optical fibers have a wider band width (compared to conventional copper cables). This helps in transmitting the information data on a single line and at very fast rates. [10^{14} bps as compared to about 10^4 bps in ordinary communication line].
2. No Electromagnetic interference: Electromagnetic interference and disturbance in the transmission is a very common phenomenon in ordinary copper cables. However, the optical fiber cables are free from electromagnetic interference.
3. Low attenuation: Compared to metallic cables, optical fibers have a low attenuation level. The loss in optical fibers is very low of the order of 0.1 to 0.5 dB/km.
4. No Electrical Hazards: Since optical fibers carry information in the form of optical signals, there are no problems such as short – circuiting, shock etc.
5. High Security: Unlike electrical transmission lines, there is no signal radiation around the optical fiber. Hence the transmission is secure. The tapping of the light wave is ruled out, if done leads to loss of signal and can easily be detected.
6. Flexibility: Optical fiber cables are flexible compared to copper cables so that it can take any zig- zag shape. Also they are small in size, lightweight & have long life.

Disadvantages of Optical Communication:

1. Bend losses: Bend losses are more in optical fibers while in copper cables bending has no effect in signal strength.
2. Cable splicing: Splicing is the method of joining two optical fibers. Effective splicing operation costs both time and money.
3. Thermal stability: The expansion and contraction of optical fibers under varying environmental conditions affect the alignment at the splices and cause transmission loss.
4. Cost: Compared to conventional cables, the initial cost of optical fibers is high and also its maintenance is time consuming and expensive compared to a conventional copper cable.